

Lecture 3.1: Theory of Random Number Generation

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Motivation

Every simulation — a Monte Carlo integral, a bootstrap resample, a randomized algorithm — ultimately needs a long supply of numbers that *behave* like independent draws from the $\text{Uniform}(0, 1)$ distribution. A computer cannot produce numbers that are genuinely random (its arithmetic is entirely deterministic), so instead it produces numbers that are deterministic but designed to be statistically indistinguishable from random ones. Understanding exactly how this is done, and exactly where it can go wrong, is the purpose of this lecture.

Intuition

Imagine a clock face with m marks on it. Starting at some mark X_0 , suppose a rule says: multiply the current mark by a , add c , and read off the remainder after dividing by m — that remainder becomes the next mark. Even though this rule is completely mechanical, the sequence of marks visited can look erratic, jumping around the clock face in no obvious pattern, *provided* a , c , and m are chosen well. If they are chosen badly, the sequence might visit only a handful of marks before repeating itself, which would be disastrous for a simulation expecting thousands of “independent” values. The mathematics below answers, precisely, which choices of a, c, m guarantee that *every* mark on the clock face is visited exactly once before the sequence repeats.

Pseudo-Random Numbers

Definition 1.1. A *pseudo-random number generator* (PRNG) is a deterministic recurrence $X_{n+1} = g(X_n)$ on a finite state space $S = \{0, 1, \dots, m-1\}$, together with a seed $X_0 \in S$ and an output map $U_n = X_n/m \in [0, 1)$, intended to imitate independent $\text{Uniform}(0, 1)$ draws. Because S has only m elements, the sequence (X_n) must eventually repeat itself (among any $m + 1$ terms, two must coincide); the smallest P with $X_{n+P} = X_n$ for all large n is called the *period*. If $P = m$, every possible state is visited once per cycle, and the generator is said to have **full period** — the best a deterministic recurrence on m states can achieve.

The Linear Congruential Generator

Definition 1.2. The *linear congruential generator* (LCG) is the clock-arithmetic rule described above,

$$X_{n+1} = (aX_n + c) \bmod m,$$

with multiplier a , increment c , and modulus m . Setting $c = 0$ gives the *multiplicative* generator (MCG), historically the first version proposed (Lehmer, 1949); keeping $c \neq 0$ gives the *mixed* generator, which — as the next theorem shows — can achieve full period on *any* modulus, not only a prime one.

The following theorem is the central result of this lecture: it converts the vague hope “the sequence should look random” into three checkable arithmetic conditions on a , c , and m .

Theorem 1.1 (Hull–Dobell, 1962). The LCG has full period m for every seed if and only if

1. $\gcd(c, m) = 1$;
2. $a \equiv 1 \pmod{p}$ for every prime $p \mid m$;
3. $a \equiv 1 \pmod{4}$ if $4 \mid m$.

Proof. The first step is to write down the state after n steps explicitly, rather than one step at a time. Unrolling the recurrence $X_1 = (aX_0 + c) \bmod m$, $X_2 = (aX_1 + c) \bmod m, \dots$ and substituting repeatedly gives the closed form

$$X_n = (a^n X_0 + cS(n)) \bmod m, \quad S(n) = \sum_{i=0}^{n-1} a^i,$$

which can be checked by induction: it holds for $n = 1$, and if it holds for n then $X_{n+1} = (aX_n + c) \bmod m = (a^{n+1}X_0 + c(aS(n) + 1)) \bmod m$, and $aS(n) + 1 = S(n+1)$ by direct expansion of the geometric sum.

Full period means that $n = m$ is the *smallest* positive integer for which X_n returns to X_0 , and this must hold *simultaneously for every possible seed* X_0 — not just for one lucky starting value. Comparing the closed form above at two different seeds (or, equally, comparing the coefficient of X_0 and the constant term separately) shows that “ $X_n \equiv X_0 \pmod{m}$ for every X_0 ” is equivalent to the pair of conditions $a^n \equiv 1 \pmod{m}$ and $cS(n) \equiv 0 \pmod{m}$ holding together, and full period asks that $n = m$ be the smallest n for which both hold.

Now write $b = a - 1$. Conditions (2) and (3) say exactly that $p \mid b$ for every prime p dividing m , and that $4 \mid b$ whenever $4 \mid m$. A classical number-theoretic fact (the “lifting-the-exponent” lemma; see Knuth, 1997, §3.2.1.2) says that these two divisibility requirements on b are *precisely* what is needed to guarantee, for every prime power p^e exactly dividing m , that $p^e \mid S(n)$ as soon as $p^e \mid n$ — and no earlier. Piecing the prime-power factors of m back together with the Chinese Remainder Theorem then shows $a^m \equiv 1 \pmod{m}$ and $m \mid S(m)$, with no smaller n achieving both. Condition (1), $\gcd(c, m) = 1$, is exactly what allows $cS(n) \equiv 0 \pmod{m}$ to be replaced by the condition $m \mid S(n)$ regardless of the specific value of c . Putting these pieces together shows the smallest valid n is exactly m .

For the converse, if any one of the three conditions fails for some prime power factor of m , the map $X \mapsto (aX + c) \bmod m$, restricted to that factor, fails to be a single cycle through all its residues; since the overall cycle structure on $\mathbb{Z}_m$ is built from the cycle structures on its prime-power factors (again via the Chinese Remainder Theorem), a shorter cycle on any one factor forces a period shorter than m overall. ■

Worked example. Take $a = 5$, $c = 3$, $m = 16$, seed $X_0 = 7$. Since $16 = 2^4$, the only prime factor is $p = 2$; check $a - 1 = 4$, which is divisible by 2 (condition 2) and by 4 (condition 3, since $4 \mid 16$); also $\gcd(3, 16) = 1$ (condition 1). All three conditions hold, so the theorem guarantees full period 16. Iterating by hand,

$$\begin{aligned} X_1 &= (5 \cdot 7 + 3) \bmod 16 = 38 \bmod 16 = 6, & X_2 &= (5 \cdot 6 + 3) \bmod 16 = 33 \bmod 16 = 1, \\ X_3 &= (5 \cdot 1 + 3) \bmod 16 = 8, & X_4 &= (5 \cdot 8 + 3) \bmod 16 = 11, & X_5 &= (5 \cdot 11 + 3) \bmod 16 = 10, \end{aligned}$$

and continuing this process visits all 16 residues $0, \dots, 15$ exactly once before the sequence returns to $X_0 = 7$, exactly as promised.

Lemma 1.2. If m is prime, the MCG $X_{n+1} = aX_n \bmod m$ (seed $X_0 \neq 0$) has period equal to the multiplicative order of a modulo m ; this attains its maximum, $m - 1$, exactly when a is a **primitive root** mod m .

Proof. Because m is prime, the nonzero residues $\{1, \dots, m - 1\}$ form a group under multiplication modulo m — the group $(\mathbb{Z}/m\mathbb{Z})^*$, of order $m - 1$. The orbit $X_n = a^n X_0$ returns to its starting value X_0 exactly when $a^n \equiv 1 \pmod{m}$, that is, exactly when n is a multiple of the smallest such exponent, called the multiplicative order $\text{ord}(a)$. Lagrange’s theorem for finite groups says $\text{ord}(a)$ must divide the group’s order $m - 1$, so the period can never exceed $m - 1$; it equals $m - 1$ exactly when a ’s powers sweep out the *entire* group before repeating, which is the definition of a primitive root. ■

Worked example. With $m = 31$ (prime) and $a = 3$, computing successive powers of 3 modulo 31 — 3, 9, 27, 19, 26, 16, 17, 20, 29, 25, 13, 8, 24, 10, 30, 28, 22, 4, 12, 5, 15, 14, 11, 2, 6, 18, 23, 7, 21, 1 — shows that 1 first reappears only at the 30-th power. Since $30 = m - 1$, this confirms 3 is a primitive root modulo 31, and the corresponding MCG has the maximal possible period, 30.

Why Full Period Is Not the Whole Story

Remark. Full period only guarantees that every state is visited once per cycle; it says nothing about whether *consecutive* outputs look independent of one another when viewed several at a time. The generator RANDU ($a = 65539$, $c = 0$, $m = 2^{31}$), once used widely on IBM mainframes, illustrates this dramatically. Because $a = 2^{16} + 3$, squaring gives $a^2 = 2^{32} + 6 \cdot 2^{16} + 9$, and since $2^{32} \equiv 0 \pmod{2^{31}}$, this reduces to the exact identity $a^2 \equiv 6a - 9 \pmod{2^{31}}$. Multiplying through by X_n (recall the generator is $X_{n+1} = aX_n \pmod{2^{31}}$, so $X_{n+2} = a^2X_n \pmod{2^{31}}$) gives

$$X_{n+2} \equiv 6X_{n+1} - 9X_n \pmod{2^{31}}.$$

Numerical check. With $a = 65539$, direct computation gives $a^2 \pmod{2^{31}} = 393225$ and $6a - 9 = 6(65539) - 9 = 393225$ — the two sides agree exactly, confirming the identity is not an approximation but an exact arithmetic fact about this specific multiplier.

In words: every *third* value in the sequence is an exact linear combination of the two before it. Geometrically, this forces every triple of consecutive outputs (U_n, U_{n+1}, U_{n+2}) , plotted as a point in the unit cube, to lie on one of only 15 parallel planes, rather than filling the cube — a severe defect that is completely invisible if one only looks at pairs of consecutive values, as the figure below shows.

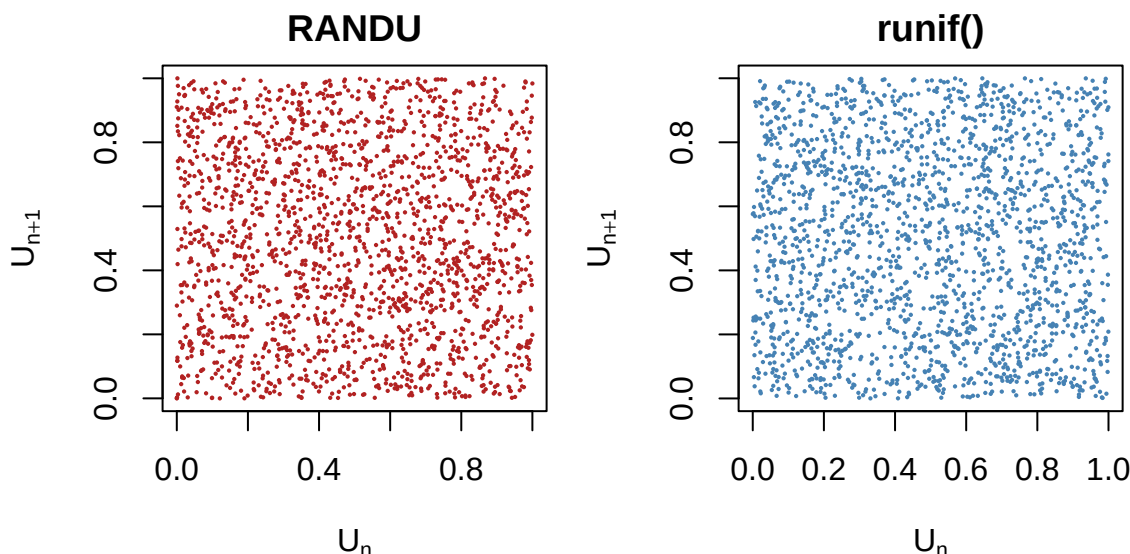


Figure 1: Successive pairs from RANDU (left) versus a well-chosen generator (right); the planar defect of RANDU only becomes visible in three dimensions.

The lesson is that the Hull–Dobell theorem and Lemma 1.2 certify only the *arithmetic* property of full period; whether the output also *looks* statistically random — the subject of the next lecture — must be checked separately.

References

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